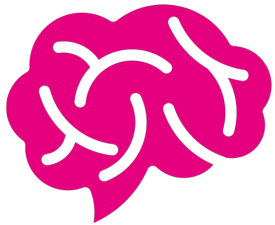


Concentration of Solutions

Lesson objectives

- Compare concentrations of solutions.
- Use particle theory to explain the difference between a dilute and concentrated solution
- Prepare solutions of different concentrations.



When you add sugar to a cup of tea what happens...

- to the sugar?
- to the mass of the tea?

What is the solute, solvent and solution in this example?

Success criteria

Lesson Title	Working towards	Working at	Working beyond
Dissolving	I can identify a solute, solvent, and solution in a mixture.	I can measure mass/volume and describe dissolving using the particle model.	I can explain how particles interact in a solution and relate solute-solvent relationships
Concentration of Solutions	I can compare how concentrated two solutions are.	I can prepare solutions of different concentrations and explain dilute vs concentrated using particle theory.	I can explain how particle theory accounts for concentration differences and evaluate methods for preparing accurate solutions.
Solubility in water	I can identify when a solution is saturated.	I can explain how concentration affects dissolving and investigate the solubility of different solutes.	I can interpret solubility data to explain saturation and compare the solubilities of different substances.
Solubility Investigation (Temperature)	I can collect data and record it in a table.	I can plan an investigation, collect accurate data, and draw a graph.	I can interpret my graph to explain how temperature affects solubility and evaluate the quality of my data.

Let's
Recap

Dissolving



8:00

Use the following words to fill the gaps in the paragraph below:

solution

nail varnish remover

solvent

soluble

insoluble

solute

When a solid dissolves in a liquid it makes a special type of mixture called a _____. The solid part of the solution is called the _____ and the liquid it dissolves in is called the _____.

The most common type of solvent is water, but some solutes dissolve in other solvents such as oil or _____. A substance which dissolves in water is called _____, a substance which does not dissolve easily is called _____.

Let's Particle diagrams

Recap

8:00

1. Draw a particle diagram to show instant coffee
2. Draw a particle diagram to show liquid hot water
3. Draw a particle diagram to show the coffee dissolved in the hot water

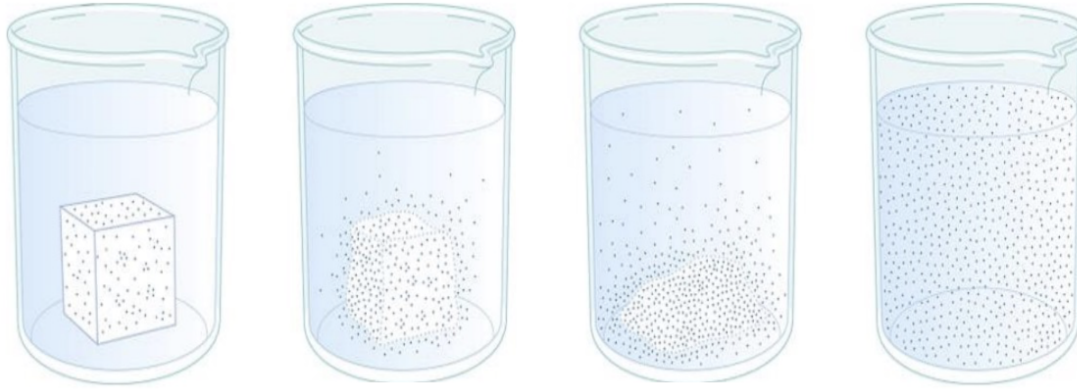


Success criteria for drawing particle diagrams:

- Draw particles that are the same size (not too small)
- Take care and draw carefully
- Full circles
- Think about the arrangement
- You can show particles to be different by either drawing them a different size or different colour

Stretch: Write a sentence to explain the arrangement of particles in each diagram

Dissolving and melting



In your own words,
what is the
difference between
dissolving and
melting?



2

1

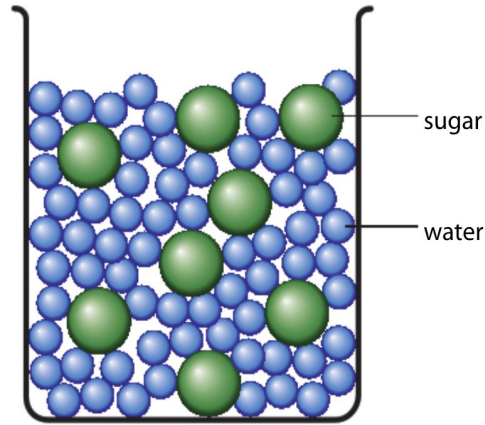
Concentration

Where have you heard the word 'concentration' before?

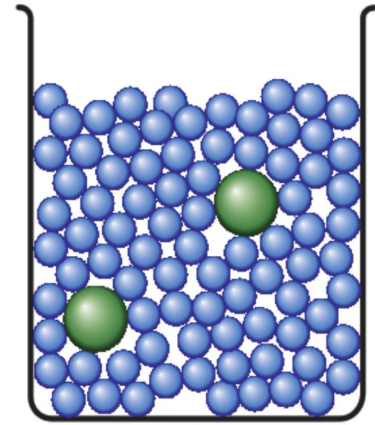


Solutions

The concentration of a solution is a **measure of the amount of solute that has been dissolved in a given amount of solvent** or solution.



A **concentrated solution** is one that has a relatively **large** amount of dissolved solute.



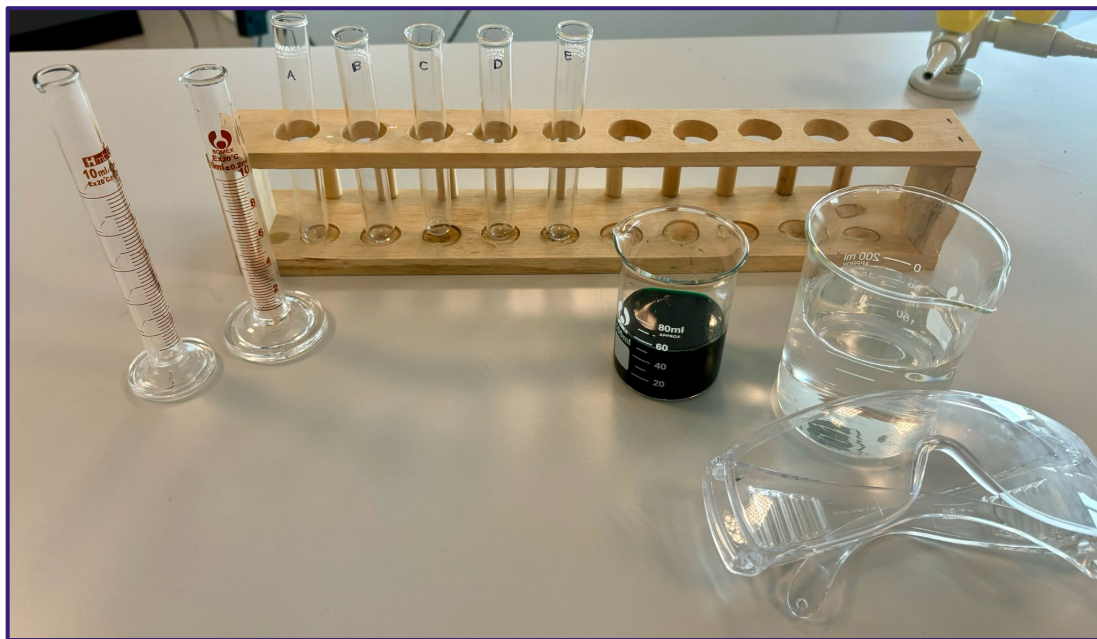
A **dilute solution** is one that has a relatively **small** amount of dissolved solute.

Making different concentrations of a solution

You will need:

- safety glasses
- test tubes
- test tube rack
- pipette
- two measuring cylinders, suitable for measuring 10 cm^3
- concentrated solution of food dye
- beaker of water

**Learner's
Book, page
46-47**



Method

- 1 Carefully measure out 10 cm^3 of the concentrated food dye solution. When you have added about the correct volume you can use the pipette to add or remove the final amount drop-by-drop, so that your measurement is as accurate as possible. Place it in a test tube and leave it in the test tube rack. This is solution A.
- 2 Carefully measure another 8 cm^3 of the concentrated food dye solution. Pour it into a test tube.
- 3 Measure out 2 cm^3 water and add it to the 8 cm^3 of food dye. Leave it in the test tube rack. This is solution B.
- 4 Use the table below to make up food dye solutions C, D and E. Place the solutions in the test tube rack, in order, from A–E.

Solution	Volume of concentrated food dye solution in cm^3	Volume of water in cm^3	Total volume in cm^3
A	10	0	10
B	8	2	10
C	6	4	10
D	4	6	10
E	2	8	10

- 5 Look carefully at the solutions you have made.

Questions

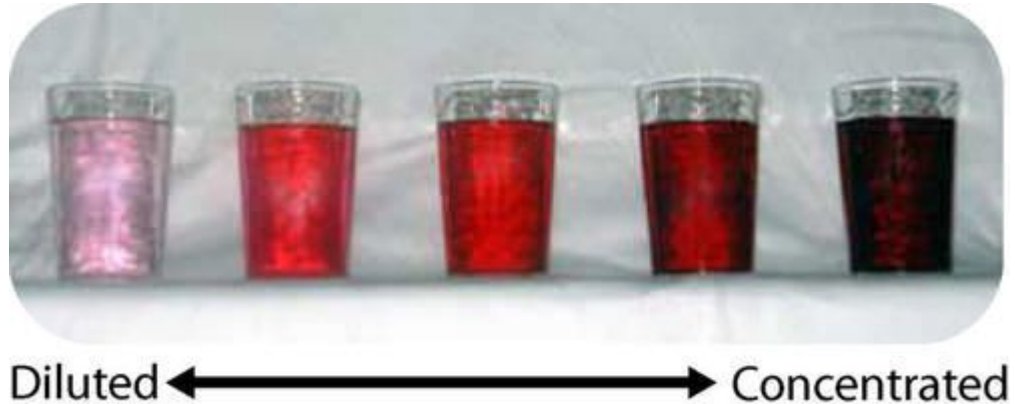
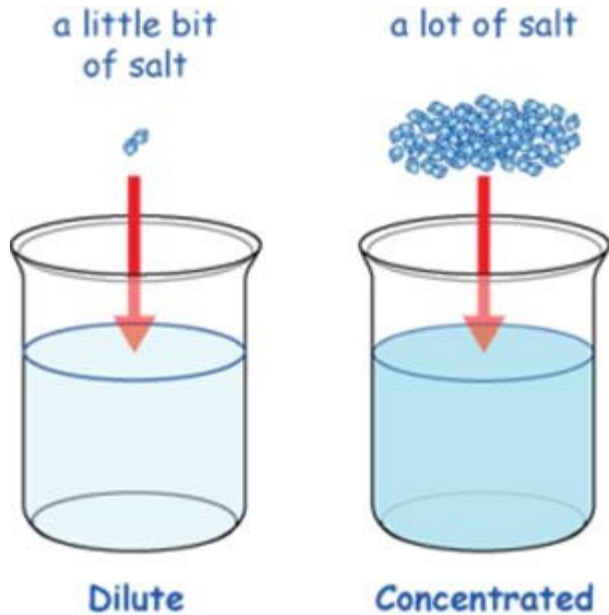
- 1 What do you notice about the solutions?
- 2 How can you tell which is the least concentrated?
- 3 If you repeated this task using a salt or sugar solution, would you be able to identify the most and least concentrated solutions? Explain your answer.
- 4 Why is it important to measure the food dye solution and the water accurately?
- 5 If you only had a measuring cylinder that measured up to 100 cm^3 , would using these same volumes of copper sulfate and water be accurate?
- 6 Compare the number of particles of food dye in the most concentrated solution of food dye and the most dilute solution.



Stretch:

Can you draw a **particle diagram** to show the solute dissolved in the solvent for solution **B** and solution **E**?

Concentration



Before you go

Explain the difference between...

a **concentrated** solution and a **less concentrated** one

You must use the words: “**solvent**”, “**solute**” and “**particle**”